

Progress Test 3.A — Elec-S1
 3.A · Kirchhoff’s laws (combinations, divider, Millman)
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Equivalent resistance (1 mark)

Give the equivalent resistance of $R_1 = 6.0\text{ k}\Omega$ and $R_2 = 3.0\text{ k}\Omega$ connected in parallel.

[Empty grid for answer]

Marking

①
 0 1
 [] []

Exercise 2. Kirchhoff’s laws (5 marks)

State Kirchhoff’s node law and mesh law. For each law, give the fundamental principle it expresses.

[Empty grid for answer]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Quick circuit laws

(3 marks)

$R_1 = 1.0\text{ k}\Omega$ and $R_2 = 3.0\text{ k}\Omega$ are in series under $u = 12\text{ V}$. The voltage u_2 across R_2 is:

- ☐ 3.0 V ☐ 4.0 V ☐ 9.0 V ☐ 12 V

A circuit has $N = 3$ electrically distinct nodes and $B = 5$ branches. The number of independent mesh equations is:

- ☐ 2 ☐ 3 ☐ 5 ☐ 8

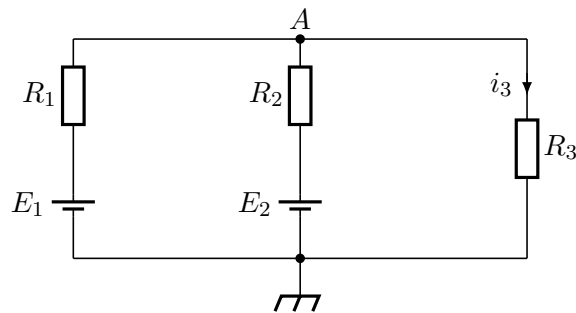
Three identical $30\ \Omega$ resistors in parallel are equivalent to:

- ☐ $90\,\Omega$ ☐ $30\,\Omega$ ☐ $10\,\Omega$ ☐ $3.3\,\Omega$

Exercise 4. Millman's theorem

(5 marks)

In the circuit below, the ground (bottom wire) is the reference of potentials. Using Millman's theorem, determine the potential V_A of node A . Deduce the current i_3 that flows through R_3 towards the ground.



$E_1 = 12\text{ V}$	$R_1 = 2.0\ \Omega$	$E_2 = 6.0\text{ V}$	$R_2 = 3.0\ \Omega$	$R_3 = 6.0\ \Omega$

Marking

1

01

2

01

3

01

4

01

5

01

— End of the test —